DO NOW — RECALL

Thinking concurrently: independent tasks

Name: _____ Class: _____ Date: _____

1. What does the MAR contain during a memory access?

2. When is A AND B true?

3. What distinguishes iterative development from a fixed sequence?

4. Search for 13 in sorted indexes 0–14. Trace the midpoint indexes inspected by standard integer midpoint selection.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The address of the memory location being accessed.

2. Model answer

Only when both operands are true.

3. Model answer

Incremental working versions and repeated feedback.

4. Model answer

low=0 high=14 → mid=7; target greater → low=8 high=14 mid=11; greater → low=12 high=14 mid=13; found. Midpoints 7,11,13.

Vocabulary — know these terms

concurrency — Tasks making progress over overlapping intervals.

dependency — A task needs another result first.

critical path — Longest dependent chain determining ideal completion time.

race condition — Timing-dependent outcome caused by unsafe shared access.

synchronisation — Coordination of operations using shared state.

NEW LEARNING

Read, annotate and explain

Find tasks without dependencies

Dataset download and image processing can operate concurrently if they use separate resources.

Respect the final dependency

Report assembly cannot start until BOTH preparation tasks finish.

Calculate the ideal gain

Sequential 9 minutes, parallel 6 minutes. Actual performance depends on scheduling and resource contention.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Two independent data checks take 5 and 3 minutes; publishing takes 2 more. Calculate ideal sequential and concurrent times.

Predict the method

Add all durations for the sequential path: 10 minutes.

Check and explain

Run independent checks together, wait 5 minutes, then publish in 2: total 7 minutes.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Tasks A (3 min) and B (5 min) are independent, then final C (2 min). Find sequential and ideal parallel completion times.

Question 2

Two withdrawals of £10 and £20 both read the same initial £100 balance before either writes an update. Explain a possible lost update and propose a safeguard.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Four tasks: parse A=4 min, fetch B=6 min, validate C=2 min (requires A and B), publish D=1 min (requires C). Find shortest ideal time and explain two constraints.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

All parts of an algorithm can always run concurrently if enough cores are available.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.